

Sakshi Sanjay Khade

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Education

Arizona State University (ASU) 05/2026
Master of Science in Robotics and Autonomous Systems (Artificial Intelligence) GPA: 3.81
Relevant Coursework: Machine Learning Acceleration, Artificial Intelligence, Real-Time Embedded Systems

Terna Engineering College, India 06/2024
Bachelor of Engineering in Information Technology GPA: 3.32
Relevant Coursework: Artificial Intelligence and Data Science, Internet of Things, AR-VR, Big Data Analytics

Technical Skills

Programming & Scripting: Python (NumPy, Pandas, TensorFlow), C, C++, JavaScript, React.js, Node.js, MySQL, Lingua Franca

Tools & Platforms: Firebase, Git, AWS, Tableau, Power BI

Technical Expertise: MATLAB, Data Processing & Visualization, BLE Sensor Integration

Productivity: Microsoft Office Suite (Advanced), Google Workspace, Figma, Adobe Suite

Professional Work Experience

Graduate Student Assistant, ASU — CSE 522: Real-time Embedded Systems (PT) Aug 2025 – Dec 2025

- Guiding a cohort of 52 graduate students in robotics labs using Lingua Franca and C, simplifying complex real-time embedded systems concepts.
- Designing and grading homework and lab assignments, providing actionable feedback to enhance learning.
- Conducting flipped learning sessions with live coding demos, fostering engagement and practical understanding.
- Collaborating with faculty to refine course materials and lab exercises for smoother delivery.

Data Analytics Intern, SRRS Software Solutions Pvt. Ltd., India Mar 2024 – Aug 2024

- Automated data cleaning for 50K+ sales records using Python and MySQL, improving accuracy by 30%.
- Built and maintained 5+ dashboards in Power BI and Tableau to track revenue trends.
- Developed forecasting models in Python and Excel, achieving 85% accuracy in quarterly projections.
- Documented workflows and collaborated with teams to translate insights into business strategy.

Front-End Developer Intern, The Language Network, India Oct 2023 – Feb 2024

- Delivered responsive, mobile-friendly UIs with HTML/CSS/JavaScript, reducing load times by 35%.
- Integrated Git-based version control and supported deployment of student dashboards.
- Resolved UI/UX issues across devices by collaborating with cross-functional teams.

Projects

Real-Time Traffic Data Pipeline & Forecasting Present

- Building a Spark-based data pipeline to process and transform 20K+ daily traffic records across 5 locations.
- Achieving 88% prediction accuracy using Prophet for time-series forecasting.
- Developing automated ETL workflows and dashboards in Power BI for real-time monitoring.
- Deploying the pipeline on cloud platforms to enhance scalability, real-time availability, and cost efficiency.

Emotion Classification — Excitement vs Fear from Physiological Data May 2025 – Present

- Designed a real-time emotion recognition system using Random Forest, achieving 96% accuracy.
- Implemented deep learning models (CNN, LSTMs) for enhanced classification accuracy in detecting emotional states.
- Applied advanced signal processing and feature engineering on multimodal physiological data.
- Integrated BLE-based wearable sensors (Empatica E4, Rhythm24) into a custom Android pipeline for testing.

Twitter Sentiment Analysis for Brand Monitoring Dec 2024

- Processed 50K+ tweets/day with Twitter API and NLP pipelines in Python.
- Built LSTM and Transformer-based classifiers (Hugging Face) with 93% accuracy.
- Designed an end-to-end analytics pipeline, visualizing sentiment trends in Tableau for real-time insights.

AI-Powered Emotion Detection & Intelligence Platform

Present

- Developed a full-stack AI platform for real-time emotion analysis from text, integrating sentiment, valence, and arousal metrics with an adaptive conversational agent.
- Architected scalable backend services using FastAPI deployed on AWS EC2 with PostgreSQL, ensuring low-latency inference and secure, high-availability APIs.
- Built interactive dashboards in React to visualize emotional trends, session history, and engagement analytics for improved interpretability.

Quit Hero — Addiction Recovery Platform

Jan 2026

- Built a full-stack addiction recovery platform with a mobile-first React/TypeScript UI and PocketBase backend, featuring structured programs, craving support, journaling, and achievement tracking.
- Implemented a multi-step KYC onboarding flow with personalization, quit-date tracking, reminders, and a RESTful architecture using React Query for caching, offline support, and error handling.
- Developed an admin dashboard with role-based access control, analytics, support ticketing, and real-time progress visualizations using Recharts.

Rave — Event Ticketing Platform

Nov 2025

- Engineered a full-stack ticketing platform using Next.js and Node.js with multi-currency support and organizer dashboards.
- Integrated Razorpay for payments with automated refunds, GST-aware T+2 settlements, and secure authentication via PocketBase and Google OAuth with role-based access control.
- Implemented real-time seat inventory, QR-based check-in, and D3.js analytics dashboards, deployed using Docker on AWS-ready infrastructure.

Restaurant Management System — Multi-Tenant Platform

Dec 2025

- Built a multi-tenant restaurant management system in a TurboRepo monorepo with customer ordering, back-office dashboard, and marketing site, supporting QR-based table ordering and Razorpay payments.
- Developed interactive floor-plan management with live table states and a Kitchen Display System (KDS) for real-time order tracking.
- Implemented RBAC (Admin / Manager / Staff), tenant isolation using PocketBase, real-time subscriptions, reservations, and daily sales and GST reporting features.